# Yakeen Plus (2025)

## SHORT PRACTICE TEST - 01

**DURATION: 60 Minutes** | **DATE : 11/09/2024** | **M.MARKS : 192**

## Topics Covered

| | |
|---|---|
| **Physics:** | Basic Maths & Calculus (Mathematical Tools), Vectors Units and Measurements |
| **Chemistry:** | Some Basic Concept of Chemistry, Redox Reaction |
| **Biology:** | **(Botany):** Cell: The Unit of Life<br>**(Zoology):** Structural Organization in Animals |

**General Instructions:**

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **60 minutes** duration.
3. The test booklet consists of **48** questions. The maximum marks are **192**.
4. All questions are compulsory.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

**OMR Instructions:**

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

**Name of the Student (In CAPITALS) :** ______________________

**Roll Number :** ______________________

**OMR Bar Code Number :** ______________________

**Candidate's Signature :** ______________ **Invigilator's Signature** ______________

# SECTION-(I) PHYSICS

**1.** Find sum of infinite term

$1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+.......$

(1) $\frac{1}{2}$ (2) 1
(3) 2 (4) $\frac{3}{2}$

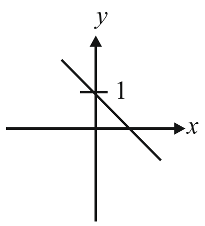
**2.** The greatest value of the function 8 sinθ – 6cosθ is:

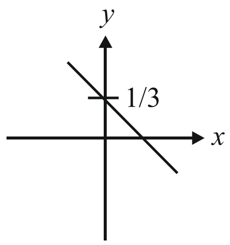
(1) 10 (2) 12
(3) 20 (4) 15

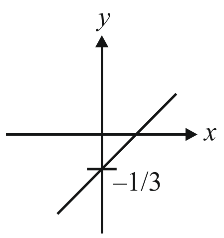
**3.** Which of the following combinations of three dimensionally different physical quantities $P$, $Q$, $R$ can never be a meaningful quantity?

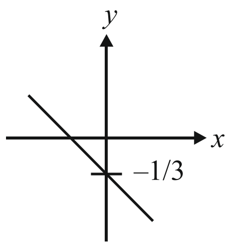
(1) $PQ-R$ (2) $\frac{PQ}{R}$
(3) $\frac{P-Q}{R}$ (4) $\frac{PR-Q^2}{QR}$

**4.** Correct graph of $3x - 3y - 1 = 0$ is;

(1) 


(2) 


(3) 


(4) 


**5.** $\log_3 x^2 = 4$, find the value of $x$;

(1) 3 (2) 5
(3) 7 (4) 9

**6.** Cos2A is equal to:

(1) $1-2\sin^2 A$ (2) $2\cos^2 A - 1$
(3) $\cos^2 A - \sin^2 A$ (4) All of these

**7.** $\int 3x^2 dx$

(1) $x^3 + C$ (2) $6x + C$
(3) $2x^2 + C$ (4) $x^2 + C$

**8.** $\frac{d}{dx}\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2$ is equal to;

(1) $1+\frac{1}{x^2}$ (2) $-1+\frac{1}{x^2}$
(3) $1-\frac{1}{x^2}$ (4) $x^2-1$

**9.** If $\left|\vec{A}+\vec{B}\right|=\left|\vec{A}-\vec{B}\right|$, then what will be angle between $\vec{A}$ and $\vec{B}$? ($\vec{A}$ and $\vec{B}$ are non zero vectors)

(1) 30° (2) 45°
(3) 60° (4) 90°

**10.** Equation $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, shows:

(1) straight line
(2) circle
(3) ellipse
(4) hyperbola

**11.** Two forces of 12 N and 8 N act upon a body. The resultant force on the body has maximum value of:

(1) 4 N (2) 0 N
(3) 20 N (4) 8 N

**12.** The dimensions of $\frac{a}{b}$ in the equation, where $P$ is pressure $P=\frac{a-t^2}{bx}$, $x$ is distance and $t$ time are:

(1) $[MT^{-2}]$
(2) $[M^2LT^{-3}]$
(3) $[ML^3T^{-1}]$
(4) $[LT^{-3}]$

# SECTION-(II) CHEMISTRY

**13.** The oxidation number of Cr in $K_2Cr_2O_7$ is;
(1) +6 (2) –7
(3) +7 (4) –2

**14.** Empirical formula of a compound is $CH_2O$ and its vapour density is 30. Molecular formula of the compound is;
(1) $C_3H_6O_3$
(2) $C_2H_4O_2$
(3) $C_2H_4O$
(4) $CH_2O$

**15.** $P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$ is an example of;
(1) Intermolecular redox reaction
(2) Intramolecular redox reaction
(3) Disproportionation redox reaction
(4) Combination redox reaction

**16.** For the reaction,
$MnO_4^- + C_2O_4^{-2} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$
The **correct** coefficient of the reactants for the balanced reaction are;

| | $MnO_4^-$ | $C_2O_4^{2-}$ | $H^+$ |
|---|---|---|---|
| (1) | 2 | 5 | 16 |
| (2) | 16 | 5 | 2 |
| (3) | 5 | 16 | 2 |
| (4) | 2 | 16 | 5 |

**17.** Which of the following has the highest mass?
(1) 1 g-atom of phosphorous
(2) 2 moles of water
(3) 22.4 L of $CO_2$ gas at STP
(4) $6.02 \times 10^{23}$ atoms of sulphur

**18.** For a gaseous reaction,
$Cl_2(g) + PCl_3(g) \rightarrow PCl_5(g)$
If 10 mL of $Cl_2$ completely react with $PCl_3$ then volume of $PCl_5$ formed will be;
(1) 10 ml (2) 20 ml
(3) 5 ml (4) 1 ml

**19.** Assuming chlorine element has two isotopes as $Cl^{35}$ and $Cl^{37}$ with their percentage abundance as 25% and 75% respectively. The average atomic weight of Cl is;
(1) 35.5 u
(2) 40 u
(3) 36.5 u
(4) 38.5 u

**20.** 12 g of Mg (molar mass = 24 g/mol) on reacting completely with HCl gives hydrogen gas, the volume of which at STP would be:
$Mg + 2HCl \rightarrow MgCl_2 + H_2$
(1) 22.4 L (2) 11.2 L
(3) 44.8 L (4) 6.1 L

**21.** Given below are two statements:
**Statement I:** The assigned oxidation state of oxygen in $O_2F_2$ is +1.
**Statement II:** Oxidation state of oxygen in $KO_2$ is –2.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

**22.** The mass of sodium acetate ($CH_3COONa$) required to prepare 250 mL of 0.35M aqueous solution is; (Molar mass of $CH_3COONa$ is 82 g $mol^{-1}$)
(1) 6.11 g (2) 7.18 g
(3) 8.64 g (4) 9.25 g

**23.** Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
**Assertion A:** $ClO_4^-$ can **not** show disproportionation reaction.
**Reason R:** Minimum oxidation state of Cl in any compound is –1.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

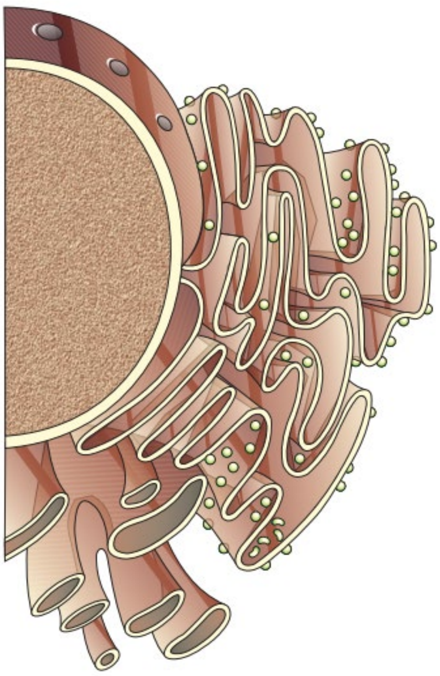
**24.** Volume of 3 M NaOH (Molar mass = 40 g $mol^{-1}$) which can be prepared from 84 g of NaOH is;
(1) 0.8 L (2) 0.7 L
(3) 0.9 L (4) 0.1 L

## SECTION-(III) BOTANY

**25.** The cylindrical structures that has cart wheel appearance:
(1) are made up of nine evenly spaced peripheral fibrils of tubulin protein.
(2) are made up of nine peripheral doublets and two central singlet fibrils.
(3) has central tubules connected by bridges and is enclosed by a central sheath.
(4) consists of a central hub and nine radially arranged spokes made of nucleoprotein fibres.

**26.** Which of the following is **correct** about cell membrane?
A. The detailed structure of cell membrane was studied after the advent of light microscopy.
B. The major lipids are phospholipids that are arranged within the membrane.
C. Depending upon ease of extraction, membrane proteins can be classified as integral and peripheral.
D. The fluid mosaic model was proposed by Singer and Nicolson.
E. The membrane of erythrocyte has approximately 60-70 percent protein and 40 percent lipids.
Choose the most appropriate answer from the options given below:
(1) B, C, D and E only
(2) B, C and D only
(3) A B, C, and D only
(4) All of these

**27.** How many statements of the following are **true** regarding the figure given below.



A. It forms network of tiny tubular structures scattered in the cytoplasm.
B. Rough endoplasmic reticulum is frequently observed in the cells actively involved in protein synthesis.
C. It consist of many flat, disc-shaped sacs or cisternae of 0.5μm to 1.0μm diameter.
D. It divides the intracellular space into two distinct compartments.
E. It is extensive and continuous with the outer membrane of the nucleus.
(1) Two (2) Five
(3) Three (4) Four

**28.** Consider the following statement and select **correct** option having all **correct** statement.
A. The vacuole is bound by a double membrane called tonoplast.
B. The outer and inner membrane of mitochondria has its own specific enzyme.
C. Ribosomes are composed of ribonucleic acid (RNA) and proteins and are surrounded by single membrane.
D. Cytoskeleton are involved in many functions such as mechanical support and motility.
E. The content of nucleolus is continuous with cytoplasm.
F. Microbodies contain various enzymes, are present in both plant and animal cells.
G. Prokaryotic flagella are structurally similar to eukaryotic flagella.
Choose the most appropriate answer from the options given below:
(1) B, D and F only
(2) A, C, D, E and F only
(3) A, B, D and G only
(4) A, C, E and G only

**29.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
**Assertion A:** The number of mitochondria per cell depends on the physiological activity of the cells.
**Reason R:** Mitochondria produce cellular energy in the form of ATP.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

**30.** Which of the given statement is **wrong**.

(1) Steroidal hormones are synthesised in smooth endoplasmic reticulum.
(2) Golgi apparatus performs the function of packaging of material.
(3) Carbohydrases are found in lysosome that are function at acidic pH.
(4) Chromatin stained by acidic dyes.

**31.** How many of the following is/are **incorrect**?

A. Sodium-potassium pumps move across cell membrane by passive transport.
B. The quasi-fluid nature of lipid enables lateral movement of proteins within lipid bilayer.
C. Polar solutes require carrier protein to facilitate their transport across the membrane called facilitated diffusion.
D. Both facilitated diffusion and active transport are energy dependent processes.
E. Neutral solutes may move across the membrane by simple diffusion.

Choose the most appropriate answer from the options given below:

(1) One (2) Three
(3) Four (4) Two

**32.** Match **List-I** with **List-II.**

| List-I | | List-II | |
|---|---|---|---|
| (A) | Aleuroplasts | (I) | Oils and fats |
| (B) | Leucoplasts | (II) | Potato |
| (C) | Amyloplasts | (III) | Protein |
| (D) | Elaioplasts | (IV) | Xanthophylls |
| (E) | Chromoplasts | (V) | Colourless plastid |

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-II, D-I, E-V
(2) A-III, B-V, C-I, D-II, E-IV
(3) A-III, B-V, C-II, D-I, E-IV
(4) A-IV, B-I, C-II, D-III, E-V

**33.** How many statements are **incorrect** related to chloroplast?

A. Mainly found in mesophyll cells.
B. Length 2 - 4 μm and width 5-10 μm.
C. *Chlamydomonas* contain 20 - 40 chloroplasts per cell.
D. Membrane of thylakoid enclose a space called lumen.
E. Stroma lamella connect thylakoids of different grana.
F. Stroma contain enzymes for synthesis of carbohydrates and proteins.

(1) Three (2) Two
(3) Five (4) Four

**34.** Match **List-I** with **List-II.**

| List-I | | List-II | |
|---|---|---|---|
| (A) | Chloroplast | (I) | Contain chlorophyll and carotenoid pigments |
| (B) | Plasmodesmata | (II) | Sites of aerobic respiration |
| (C) | Mitochondria | (III) | Provides barrier to undesirable macromolecules |
| (D) | Cell wall | (IV) | Connects cytoplasm of adjacent cells |

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-II, D-I
(2) A-I, B-IV, C-II, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-III, C-IV, D-I

**35.** Mark the **wrong** statements:

A. The cell wall of bacteria determines the shape of the cell.
B. Bacteria that take up the gram stain are Gram positive.
C. Unicellular organisms are capable of independent existence.
D. Schleiden and Schwann had given final shape to cell theory.
E. Several ribosomes may attach to a single mRNA and form a chain called polyribosomes.
F. The bacterial cells, have a chemically simple cell envelope.
G. The cytoplasm is the main arena of cellular activities in both the plant and animal cells.

Choose the most appropriate answer from the options given below:

(1) A, B, E and G only
(2) B and C only
(3) D and F only
(4) A, D and F only

**36.** Match **List-I** with **List-II.**

| List-I | | List-II | |
|---|---|---|---|
| (A) | Slime layer | (I) | DNA replication |
| (B) | Capsule | (II) | Contains pigments |
| (C) | Mesosomes | (III) | Motility |
| (D) | Pili | (IV) | Conjugation |
| (E) | Fimbriae | (V) | Loose sheath |
| (F) | Chromatophores | (VI) | Bristle like |
| (G) | Flagella | (VII) | Tough |

Choose the **correct** answer from the options given below:

(1) A-VI, B-IV, C-V, D-I, E-II, F-III, G-VII
(2) A-V, B-VII, C-I, D-IV, E-VI, F-II, G-III
(3) A-II, B-III, C-VI, D-VII, E-I, F-IV, G-V
(4) A-VII, B-VI, C-II, D-III, E-V, F-I, G-IV

[Short Practice Test-01 | Yakeen Plus (2025) | 11/09/2024]
Structural Organization in Animals

# SECTION-(IV) ZOOLOGY

**37.** Match **List-I** with **List-II** w.r.t digestive system of cockroach.

| List-I | | List-II | |
|---|---|---|---|
| (A) | The structures used for storing of food. | (I) | Gizzard |
| (B) | Ring of 6-8 blind tubules at junction of foregut and midgut. | (II) | Gastric caeca |
| (C) | Ring off 100-150 yellow coloured thin filaments at junction of midgut and hindgut. | (III) | Malpighian tubules |
| (D) | The structures used for grinding the food. | (IV) | Crop |

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-III, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-II, C-IV, D-I

**38.** Given below are two statements:

**Statement I:** Ligaments are dense irregular tissue.

**Statement II:** Cartilage is a dense regular tissue.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

**39.** Which of the following is **not** a connective tissue?

(1) Adipose tissue
(2) Bone
(3) Blood
(4) Neuroglia

**40.** Tegmina in cockroach arises from:

(1) metathorax.
(2) mesothorax.
(3) prothorax.
(4) both (1) and (2)

**41.** Malpighian tubules in cockroach are used for:

(1) respiration.
(2) reproduction.
(3) excretion.
(4) digestion.

**42.** Select the **correct** route for the passage of sperms in male frogs.

(1) Testes → Vasa efferentia →Kidney → Seminal vesical → Urinogenital duct → Cloaca
(2) Testes → Vasa efferentia → Bidder's canal → Ureter → Cloaca → Kidney
(3) Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca
(4) Testes → Bidder's canal → Kidney → Vasa efferentia → Urinogenital duct → Cloaca

**43.** Function of gap junction is to:

(1) facilitate communication between adjoining cells of connecting cytoplasm for rapid transfer of ions, small molecules and some large molecules.
(2) stop substance from leaking across a tissue.
(3) separate two cells from each other.
(4) performing cementing to keep neighboring cells together.

**44.** In frogs, hindlimbs end in _____ digits and forelimbs end in ______ digits.

Choose the **correct** option to fill the blanks.

(1) five; four

(2) six; four

(3) four; five

(4) five; six

**45.** Read the following statements [**A-E**] and choose the **correct** option.

A. Bones support and protect softer tissues and organs.

B. Cuboidal epithelium is found in ducts of glands.

C. Neuroglia make up less than one half the volume of neural tissues in our body.

D. Adipose tissue is specialised to store fats.

E. Intercalated disc is the characteristic feature of smooth muscle.

(1) A, B and E only

(2) A, B and D only

(3) B, C and E only

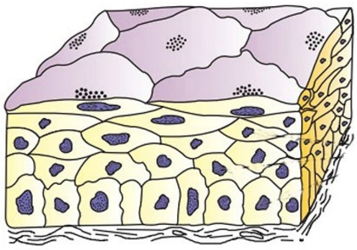
(4) A, C and D only

**46.** In frogs, cloaca is used to pass:

(1) faecal matter. (2) gametes.

(3) urine. (4) all of these

**47.** How many of the following statements is/are **correct** w.r.t the given below diagram?



A. Multilayered epithelium

B. Limited role in secretion and absorption

C. Main Function is to provide protection against chemical and mechanical stresses.

D. They cover the dry surface of skin, moist surface of buccal cavity and pharynx.

(1) Four (2) Three

(3) Two (4) One

**48.** Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Stomach and intestine of our body are lined by simple columnar epithelium.

**Reason R:** Columnar epithelium helps in secretion and absorption.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is not the correct explanation of A.